

Probability

US Common Core – High School Geometry

Domain: Conditional Probability and the Rules of Probability (S-CP.A, S-CP.B)

1 Sample Spaces and Probability

Definitions

The **sample space** of an experiment is the set of all possible outcomes. An **event** is any subset of the sample space.

Theoretical Probability

If all outcomes are equally likely,

$$P(\text{event}) = \frac{\text{number of favorable outcomes}}{\text{total number of outcomes}}.$$

Example

A standard die is rolled once. Find $P(\text{rolling a number greater than 4})$.

Favorable outcomes: $\{5, 6\}$; total outcomes: $\{1, 2, 3, 4, 5, 6\}$.

$$P(\text{roll} > 4) = \frac{2}{6} = \frac{1}{3}.$$

Complement of an Event

The **complement** of event A (written A^c or “not A ”) consists of all outcomes not in A :

$$P(A) + P(A^c) = 1.$$

2 Independent and Dependent Events

Independent Events

Two events are **independent** if the occurrence of one does not affect the probability of the other:

$$P(A \text{ and } B) = P(A) \cdot P(B).$$

Dependent Events

Two events are **dependent** if the occurrence of one *does* affect the probability of the other. In that case,

$$P(A \text{ and } B) = P(A) \cdot P(B | A),$$

where $P(B | A)$ is the **conditional probability** of B given that A has occurred.

Example (Independent)

A coin is flipped and a die is rolled. Find $P(\text{heads and a } 6)$:

$$P(\text{heads}) \cdot P(6) = \frac{1}{2} \cdot \frac{1}{6} = \frac{1}{12}.$$

Example (Dependent)

A bag has 5 red and 3 blue marbles. Two marbles are drawn *without replacement*. Find $P(\text{both red})$:

$$P(\text{red}_1) \cdot P(\text{red}_2 | \text{red}_1) = \frac{5}{8} \cdot \frac{4}{7} = \frac{20}{56} = \frac{5}{14}.$$

3 Two-Way Tables

Definition

A **two-way table** displays data classified by two categories, one along rows and one along columns, and is used to find joint, marginal, and conditional probabilities.

Example

100 students were surveyed on whether they play a sport, by grade:

	Plays a Sport	No Sport	Total
Grade 9	28	12	40
Grade 10	24	16	40
Grade 11	8	12	20
Total	60	40	100

$$P(\text{plays a sport}) = \frac{60}{100} = 0.6, \quad P(\text{Grade 9} | \text{plays a sport}) = \frac{28}{60} \approx 0.467.$$

4 Disjoint and Overlapping Events

Disjoint (Mutually Exclusive) Events

Two events cannot occur at the same time:

$$P(A \text{ or } B) = P(A) + P(B).$$

Overlapping Events

Two events can occur at the same time:

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B).$$

Example

A card is drawn from a standard 52-card deck. Find $P(\text{king or heart})$.
There are 4 kings and 13 hearts, and 1 card (the king of hearts) is both:

$$P(\text{king or heart}) = \frac{4}{52} + \frac{13}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}.$$

5 Permutations and Combinations

Factorial

For a positive integer n : $n! = n \cdot (n - 1) \cdot (n - 2) \cdots 2 \cdot 1$, with $0! = 1$.

Permutations (order matters)

The number of ways to arrange r objects chosen from a group of n distinct objects is

$${}_nP_r = \frac{n!}{(n - r)!}.$$

Combinations (order does not matter)

The number of ways to choose r objects from a group of n distinct objects, where order does not matter, is

$${}_nC_r = \binom{n}{r} = \frac{n!}{r!(n - r)!}.$$

Example (Permutation)

In how many ways can first, second, and third place be awarded among 8 runners?

$${}_8P_3 = \frac{8!}{5!} = 8 \times 7 \times 6 = 336.$$

Example (Combination)

In how many ways can a committee of 3 be chosen from 8 people?

$${}_8C_3 = \frac{8!}{3!5!} = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = 56.$$

Note: Notice ${}_8P_3 = 336$ is exactly $3! = 6$ times larger than ${}_8C_3 = 56$ – a permutation counts every different order of the same group separately, while a combination counts each group only once.